BUSTER: Adaptive Sampling for VLM-guided Unsupervised Video Anomaly Detection

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Overview. This **supplementary document** provides additional prompt details, controlled quantitative comparisons, qualitative examples, and further analysis supporting the main paper. Specifically, it includes: (i) method clarification and prompt design, (ii) additional quantitative results, (iii) additional qualitative results.

- Supplementary **A**. Method clarification and prompt design.
- Supplementary **B**. Additional quantitative comparisons.
- Supplementary **C**. Additional qualitative examples and failure cases.

A Method Clarification and Prompt Design

A.1 Supplementary Inference and Sampling Clarification

Figure 4 summarizes two implementation aspects used throughout the paper: the distinction between *offline* and *online-streaming* inference, and the design of Density-Preserving Sampling (DPS) for burst-time VLM reasoning. The currently reported main quantitative results use the *offline* configuration, namely a bidirectional recurrent student with symmetric Gaussian smoothing, and are therefore non-causal. By contrast, the *online-streaming* implementation uses a true unidirectional GRU whose hidden state and previous feature are carried across chunks, so no future context is accessed at inference time. Its hidden-state update is

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t, \tag{14}$$

where h_t is the current hidden state, h_{t-1} is the previous hidden state, $\tilde{h}_t$ is the candidate state, and z_t is the update gate. As illustrated in Fig. 4, this recurrence can be interpreted as

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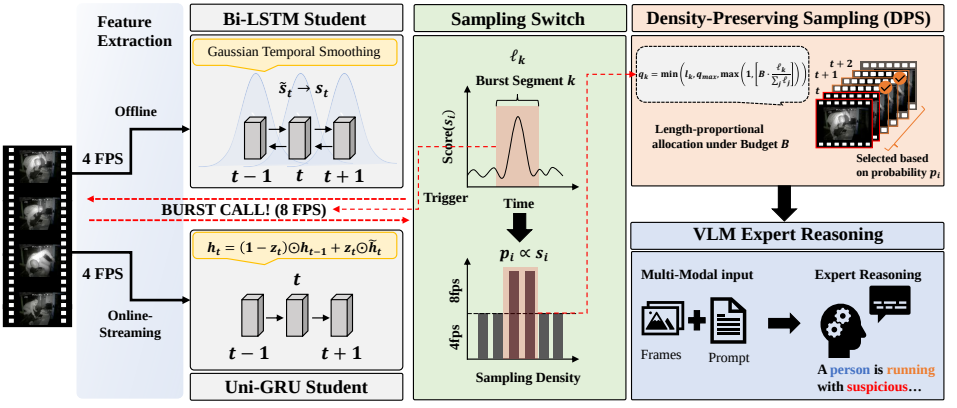


Figure 4: **Supplementary illustration of offline/online-streaming inference and Density-Preserving Sampling (DPS).** Offline scout inference uses a bidirectional student with Gaussian smoothing, whereas online-streaming inference uses a unidirectional GRU (or LSTM) with hidden-state carryover. For burst-time reasoning, DPS allocates frames across burst segments in proportion to segment length under a global budget B, subject to a per-segment cap $q_{\max}$, and then samples frames within each segment according to the student-score density. Scout features are extracted at 4 FPS, with 8 FPS re-extraction only on triggered windows.

a data-dependent EMA-like update, where z_t controls how strongly the current observation overrides the carried state.

For burst-time reasoning, DPS first allocates frames across burst segments using the length-proportional budget rule in Eq. (11) of the main paper, under the global reasoning-frame budget B and the per-segment cap $q_{\max}$. Within each burst segment, the allocated frames are then sampled according to the normalized student-score density,

$$p_i = \frac{s_i}{\sum_{j\in[i_s,i_e]} s_j}, \qquad i \in [i_s, i_e], \tag{15}$$

where s_i is the student anomaly score at index i, $[i_s, i_e]$ denotes the burst interval, and p_i is the sampling probability at position i. Thus, as shown on the right side of Fig. 4, higher-score frames are selected with higher probability, while peak-frame inclusion further preserves the most salient anomaly evidence. Together, the length-aware budget allocation and score-aware within-segment sampling retain both high-confidence cues and their surrounding temporal context for downstream VLM reasoning.

A.2 Teacher Prompt

For teacher-side pseudo-label mining, we use a binary anomaly-presence prompt and extract normalized Yes/No token probabilities from the final response logits. Our default mining mode is the logits-based formulation, which avoids free-form generation and yields a stable scalar confidence for each temporal window. Although the overall question remains binary across datasets, the prompt is adapted to the target domain and prior strength, as shown below.

Teacher Mining Prompt: UCF-Crime (logits mode, moderate prior)

System prompt
You are a surveillance video anomaly detector. Analyze frames and determine if an anomaly is present.

User prompt
Analyze these *{num_frames}* surveillance video frames.
Look for anomalies: violence, theft, accident, fire, explosion, fighting, or any suspicious activity.
Is there an anomaly in these frames? Answer with only Yes or No.

Teacher Mining Prompt: ShanghaiTech implicit (logits mode, weak contextual prior)

System prompt
You are a campus surveillance video anomaly detector. Analyze frames and determine if an abnormal activity is present in a university campus environment.

User prompt
Analyze these *{num_frames}* frames from a university campus surveillance video.
In a campus environment, normal activities include walking, standing, sitting, and talking.
Look for any abnormal or unusual behavior that deviates from typical pedestrian activities.
Is there an abnormal activity in these frames? Answer with only Yes or No.

Teacher Mining Prompt: ShanghaiTech explicit (logits mode, strong category prior)

System prompt
You are a campus surveillance video anomaly detector. Analyze frames and determine if an abnormal activity is present.

User prompt
Analyze these *{num_frames}* frames from a university campus surveillance video.
Known anomaly types in this dataset: cycling, chasing, handcart, fighting, skateboarding, vehicle intrusion, running, jumping, wandering, lifting, robbery, climbing over, throwing.
Normal activities include walking, standing, sitting, and talking.
Abnormal activities include the categories listed above.
Is there an abnormal activity in these frames? Answer with only Yes or No.

Teacher Mining Prompt: XD-Violence (logits mode, moderate domain prior)

System prompt
You are an anomaly detector for movie clips and online videos. Analyze frames and determine if a violent or abnormal event is present.

User prompt
Analyze these *{num_frames}* frames from movie clips and online videos.
Look for violent or abnormal events such as fighting, assault, accident, explosion, fire, or other suspicious activity.
Is there an anomaly in these frames? Answer with only Yes or No.

A.3 Reasoning Prompt

Unlike teacher-side pseudo-label mining, the reasoning prompt is shared across datasets. This design avoids dataset-specific reasoning templates and instead uses a unified prompt family for segment-level explanation on burst-selected evidence.

Reasoning Prompt: Anomaly Teaser (shared across datasets)

Purpose
Generate a concise segment-level explanation focused on the anomaly event.

Single-frame mode
Focus ONLY on the anomaly event and describe the most salient abnormal evidence in 2–3 sentences.

Video mode
Describe the anomaly progression in 2–4 sentences based on the provided consecutive frames.

Multi-segment mode
Provide a short anomaly-focused explanation for each burst segment separately.

Weak prior
Representative anomaly examples such as violence, theft, accident, or explosion may be mentioned as weak semantic guidance, but the prompt does not enforce a fixed closed-set label taxonomy.

Reasoning Prompt: General Analysis

This prompt requests a broader interpretation of the observed scene, including scene description, anomaly identification, and coarse risk assessment. It is used when a more general semantic analysis is required beyond anomaly-only captioning.

Reasoning Prompt: Danger Assessment

This prompt produces a coarse severity judgment, e.g., *Safe*, *Caution*, or *Dangerous*, and is used as an auxiliary interpretive output rather than a training signal.

In short, the teacher prompt is optimized for stable binary supervision, whereas the reasoning prompt is optimized for evidence-grounded segment-level explanation. A key distinction is that only the mining prompt is dataset-dependent, whereas the reasoning prompt is shared across datasets.

A.4 Why Binary Instead of Dataset-Specific Prompts?

We intentionally formulate teacher-side pseudo-label mining as a binary anomaly-presence question rather than asking the VLM to predict a predefined anomaly category. This choice is motivated by the open-set nature of surveillance anomaly detection: abnormal events vary substantially across datasets and deployment domains in terms of scene context, object appearance, human roles, clothing, lighting, and event granularity. Using a dataset-specific event prompt can inject strong category priors into the teacher and encourage the model to match a closed-set label taxonomy rather than assess abnormality from the actual visual evidence.

A binary prompt provides a more conservative but more transferable supervision signal. Instead of forcing the teacher to commit to a specific event name, it asks whether the observed segment appears anomalous at all. This is particularly important in settings where anomaly boundaries are ambiguous, where the decisive cue is visually subtle, or where similar semantic roles may appear differently across domains, for example due to country-specific uniforms, context-dependent appearances, or different severity levels of the same event type. For the same reason, we keep the reasoning prompt dataset-agnostic and use prompt adaptation only on the teacher-mining side, where lightweight prior guidance can improve sensitivity without fully constraining the teacher to a fixed closed-set event vocabulary.

A.5 Implementation Details and Hyperparameter Settings

Table 9 summarizes the default implementation and hyperparameter settings used throughout our experiments. Qwen2.5-VL-32B is used for both teacher-side pseudo-label mining and burst-time VLM reasoning, with task-specific prompts for the two stages. Importantly, the scout-to-burst threshold θ is not used to compute frame-level AUC or FAR; the student produces the detection scores independently, while θ only determines where additional temporal resolution is allocated during Scout-and-Burst inference. Offline evaluation uses bidirectional recurrence with Gaussian smoothing, whereas online evaluation operates causally with unidirectional recurrence and hidden-state carryover. Unless otherwise noted, the settings in Table 9 are fixed across datasets. Sensitivity to θ, m, and (τ_p, τ_n) is analyzed in Supplementary B.6

B Additional Quantitative Results

B.1 Deployment-Time Adaptive Sampling Analysis

We additionally validate the scout-and-burst policy under a deployment-time implementation, where scout monitoring extracts LanguageBind features directly at 4 FPS and only burst-triggered intervals are re-extracted at 8 FPS. This differs from the offline evaluation view in Eq. (10), where scout mode is represented as a 4 FPS sub-sampling of a pre-extracted 8 FPS feature stream, but yields highly similar behavior in practice.

On the UCF-Crime test set, the deployment-time adaptive implementation uses 54.4% of the 8 FPS feature budget on average, which is close to the 53.2% reported in the main paper. At the same time, it reduces the mean per-video wall-clock time from 15.63 s to 8.56 s, corresponding to a 1.90× speedup, while maintaining only 0.222% frame-level score-decision disagreement and a score MAE of 0.017 relative to fixed 8 FPS processing. These

Table 9: **Default implementation and hyperparameter settings.**

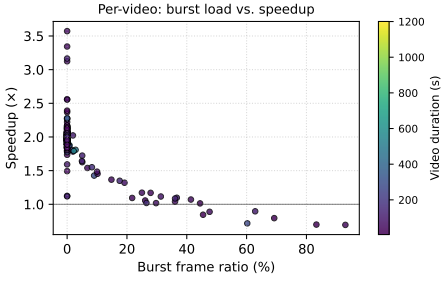
Setting	Default value
Pseudo-label mining VLM	Qwen2.5-VL-32B
Burst-time reasoning VLM	Qwen2.5-VL-32B
Student backbone	2-layer LSTM
Hidden dimension	256
Dropout	0.3
Temporal encoding	TDE (lag 1)
Student parameters	3.19M
Student FLOPs	0.32G
Ranking margin m	0.1
Positive threshold τ_p	0.5
Negative threshold τ_n	0.3
Ranking weight λ_{rank}	0.3
Smoothness weight λ_{smooth}	0.1
Scout / Burst rate	4 / 8 FPS
Scout-to-Burst threshold θ	0.5
Global reasoning budget B	16 frames
DPS per-segment cap $q_{\max}$	8 frames
Offline inference	Bi-LSTM + Gaussian smoothing ($\sigma = 16$)
Online inference	Uni-LSTM/GRU (variant-dependent)
Benchmark hardware	1× NVIDIA RTX A6000

Table 10: **Class-wise deployment-time adaptive sampling breakdown on UCF-Crime test.** Speedup is measured against fixed 8 FPS processing.

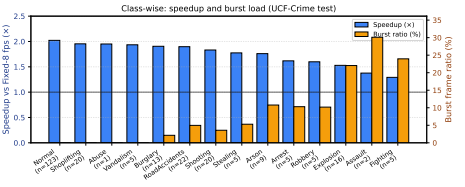
Class	N	Speedup	Data usage (%)	Burst (%)	Disagree (%)	MAE
Normal	123	×2.02	49.5	0.00	0.000	0.002
Abuse	1	×1.95	48.9	0.00	0.000	0.034
Arrest	5	×1.62	63.8	10.30	0.070	0.040
Arson	9	×1.76	62.9	10.78	1.341	0.040
Assault	2	×1.38	85.9	30.11	0.185	0.039
Burglary	13	×1.91	53.4	2.14	0.872	0.038
Explosion	16	×1.53	75.3	22.04	0.561	0.028
Fighting	5	×1.29	81.6	23.95	0.464	0.054
RoadAccidents	22	×1.90	55.2	4.97	0.075	0.031
Robbery	5	×1.60	65.7	10.18	1.536	0.038
Shooting	20	×1.83	54.7	3.58	0.484	0.035
Shoplifting	20	×1.95	49.4	0.00	0.000	0.016
Stealing	5	×1.77	58.5	5.30	0.240	0.024
Vandalism	5	×1.93	49.5	0.00	0.000	0.047
Overall	**251**	**×1.90**	**54.4**	**3.85**	**0.222**	**0.017**

results support that the offline feature-sub-sampling formulation used in the paper is a faithful proxy for actual adaptive deployment.

Table 10 and Fig. 5 further show that the speedup is strongly related to burst load. Classes with little or no burst refinement, such as *Normal*, *Shoplifting*, and *Vandalism*, achieve nearly 2× speedup, whereas classes with larger burst ratios, such as *Assault*, *Fighting*, and *Explosion*, naturally exhibit smaller gains. Overall, the adaptive implementation preserves the intended score-triggered behavior while providing substantial runtime savings.



(a) Per-video burst load vs. speedup



(b) Class-wise speedup and burst load

Figure 5: **Deployment-time adaptive sampling analysis on UCF-Crime.** (a) Per-video speedup decreases as the burst frame ratio increases, confirming that runtime gain is governed primarily by how often dense re-extraction is triggered. (b) The same trend appears at the class level: classes with little burst refinement achieve near-2× speedup, whereas classes with sustained anomalous intervals exhibit larger burst ratios and smaller gains.

B.2 Native System Comparison with Holmes-VAU

We first compare native end-to-end systems, namely the original Holmes-VAU pipeline and BUSTER with its default student backbones. This comparison is not meant to isolate individual components, since the feature encoders and scoring modules differ across systems. Rather, it provides a practical system-level reference under each method's native configuration. As shown in Table 11, BUSTER remains competitive in AUC while substantially improving AP and reducing false alarms, and it runs at a markedly higher throughput than Holmes-VAU. These results complement the controlled analysis in the main paper by showing that the practical efficiency advantage of BUSTER also holds at the native system level.

Table 11: **Native system comparison with Holmes-VAU on UCF-Crime.** This table compares each method under its native feature encoder and scoring backbone.

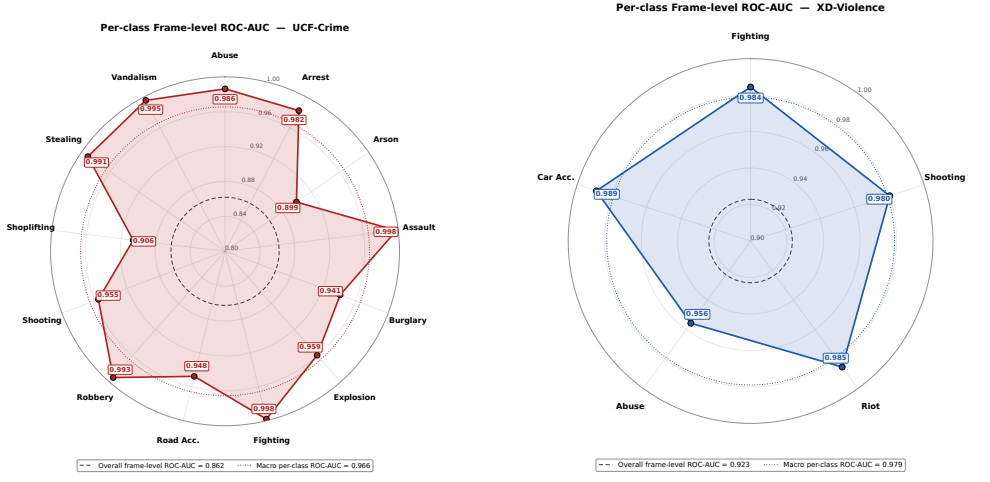
Method	Feature Encoder	Scorer	AUC (%)	AP (%)	FAR@0.1 (%)	Eff. FPS	RTF
Holmes-VAU	InternViT (1024)	URDMU (original)	86.58	28.87	2.47	167.5	5.6×
BUSTER	LanguageBind (768)	student LSTM	86.30	37.73	**0.89**	496.1	16.5×
BUSTER	LanguageBind (768)	student (GRU)	**86.68**	**38.59**	1.45	**496.3**	**16.5**×

B.3 Same-Encoder Scorer Replacement Study

To isolate the contribution of the scoring module itself, we additionally replace URDMU with the BUSTER student scorer while keeping the feature encoder fixed to InternViT. This removes the encoder difference present in the native system comparison and provides a more direct scorer-level comparison. As shown in Table 12, both the LSTM and GRU variants of the BUSTER scorer outperform the original URDMU scorer under the same InternViT features. These results indicate that the gain of BUSTER is not only due to its adaptive refinement policy, but also due to the quality of its lightweight recurrent scoring backbone.

Table 12: **Same-encoder scorer replacement study under InternViT features on UCF-Crime.** The feature encoder is fixed to InternViT, and only the scoring module is changed.

Method	Feature Encoder	Scorer	AUC (%)	AP (%)	FAR@0.1 (%)
Holmes-VAU	InternViT (1024)	URDMU (original)	86.58	28.87	2.47
Holmes-VAU	InternViT (1024)	**BUSTER LSTM**	**87.79**(+1.21)	**37.42**(+8.55)	2.11(-0.36)
Holmes-VAU	InternViT (1024)	**BUSTER GRU**	87.46(+0.88)	37.28(+8.41)	**2.10**(-0.37)



(a) UCF-Crime (b) XD-Violence

Figure 6: **Per-class frame-level ROC-AUC radar plots.** Each spoke corresponds to one anomaly class. The dashed circle indicates the overall frame-level ROC-AUC, and the dotted circle indicates the macro average over anomaly classes.

B.4 Per-Class Frame-Level AUC and Radar Plot

In Figure 6, we visualize class-wise frame-level ROC-AUC using radar plots in order to show how anomaly detection performance varies across semantic classes, beyond the single overall frame-level AUC reported in the main paper. Each spoke corresponds to one anomaly class, and its radius is the frame-level ROC-AUC computed for that class. For reference, we also overlay the overall frame-level ROC-AUC and the macro average over anomaly classes.

The class-wise AUC for class c is computed using the same frame-aligned scores as in the main paper, but restricted to a class-conditional evaluation pool:

$$\mathrm{AUC}_c = \mathrm{AUC}\left(\mathcal{F}_c^+, \mathcal{F}_c^-\right),$$

where $\mathcal{F}_c^+$ contains anomalous frames from videos of class c, and $\mathcal{F}_c^-$ contains both non-event frames from the same class videos and all frames from normal videos. The macro average is then

$$\overline{\mathrm{AUC}} = \frac{1}{K}\sum_{c=1}^{K} \mathrm{AUC}_c.$$

Because each class uses a reduced class-conditional negative pool rather than the full global negative pool, the macro per-class AUC can be higher than the overall frame-level AUC. Thus, the radar plots should be interpreted as a class-wise decomposition of frame-level detection behavior rather than as values directly comparable to the single global AUC.

Table 13: **Comparison of temporal encoding strategies for the student on UCF-Crime.**

Temporal Encoding	Input Dim	Params	AUC (%)	AP (%)	FAR@0.1 (%) ↓	Best Epoch
None (raw features)	768	3.19M	84.51	35.62	1.52	51
Positional Encoding	768	3.38M	85.20	35.73	1.32	43
Causal Conv1d (k=3)	1536	4.96M	86.19	35.38	3.46	64
Multi-lag Δ (lag 1,2)	2304	3.39M	86.21	37.36	3.01	18
TDE (Ours, lag 1)	**1536**	**3.19M**	**86.30**	**37.73**	**0.89**	**17**

In particular, the UCF-Crime radar plot shows an overall frame-level ROC-AUC of 0.862 and a macro per-class ROC-AUC of 0.966, while the XD-Violence radar plot shows 0.923 and 0.979, respectively.

B.5 Additional Analysis of Temporal Encoding

To further validate the design choice of Temporal Difference Encoding (TDE), we compare it against several alternative temporal encodings under the same student training protocol. Specifically, we evaluate: (i) no explicit temporal encoding, using only raw LanguageBind features, (ii) positional encoding, (iii) a learnable causal Conv1d temporal front-end, and (iv) a multi-lag difference encoding using first- and second-order temporal differences. All variants use the same bidirectional LSTM backbone, hidden dimension, teacher pseudo supervision, and optimization setup, so the comparison isolates the effect of the temporal encoding itself.

Table 13 shows that TDE achieves the best overall results, attaining the highest AUC and AP while also yielding the lowest FAR@0.1. Removing temporal encoding entirely causes a substantial drop in performance, confirming that temporal variation is essential for anomaly detection. A simple positional encoding improves over the raw-feature baseline, but remains clearly below TDE, suggesting that anomaly detection benefits more from explicit change cues than from sequence order alone. More expressive alternatives such as multi-lag differences and causal Conv1d approach TDE in AUC, but both incur substantially higher false alarm rates, indicating that the added temporal complexity also introduces more unstable or noisy responses. Notably, TDE achieves the strongest overall trade-off with the smallest parameter count and the fastest convergence, showing that a simple first-order temporal difference is sufficient and effective for the student scorer in this setting.

B.6 Hyperparameter Sensitivity

We further analyze the sensitivity of the Scout-to-Burst threshold θ and the ranking-related hyperparameters on UCF-Crime. As clarified in Supplementary A.5, θ does not affect how frame-level AUC or FAR are computed; it only determines where additional frames are allocated during Burst processing.

As shown in Table 14, sweeping θ from 0.30 to 0.70 changes AUC by only 0.01 percentage points, while reducing data usage from 59.3% to 51.7% and increasing refinement precision from 43.2% to 60.7%. This confirms that θ primarily controls the quality–efficiency trade-off of temporal refinement rather than the underlying detection scores.

We additionally perform a same-seed sensitivity analysis for the ranking margin m and pair-construction thresholds (τ_p, τ_n). Across the tested settings, AUC varies by at most 0.48 percentage points, although FAR is more sensitive and increases to 5.04% when lower-

confidence pair thresholds are used. We therefore retain the pre-specified default settings $m = 0.1$ and $(\tau_p, \tau_n) = (0.5, 0.3)$ across datasets for consistent evaluation.

Table 14: **Hyperparameter sensitivity on UCF-Crime.**

Setting	AUC (%)	FAR@0.1 (%)	Usage (%)	Ref. Prec. (%)
$\theta = 0.30$	86.32	0.86	59.3	43.2
$\theta = 0.50$	86.31	0.86	54.4	51.7
$\theta = 0.70$	86.31	0.86	51.7	60.7
Default m, τ	85.29	1.93	–	–
$m = 0.05$	85.70	0.60	–	–
$\tau = (0.4, 0.2)$	85.64	5.04	–	–

B.7 Direct Comparison with Dense VLM Inference

To directly assess whether dense VLM inference can replace the proposed student-based pipeline, we apply Qwen2.5-VL to full videos using the same sliding-window protocol as teacher-side pseudo-label mining.

As shown in Table 15, dense VLM inference is substantially slower and requires multiple GPUs, while still yielding lower AUC and higher FAR@0.1 than BUSTER. These results support the use of VLM supervision for offline distillation rather than continuously applying the heavy VLM over the full video stream.

Table 15: **Direct comparison with dense VLM inference on UCF-Crime.**

Method	**AUC (%)**	**FAR@0.1 (%)**	**FPS**	**RTF**
Dense VLM-32B (2 GPUs)	82.60	15.67	31.2	1.04×
Dense VLM-72B (4 GPUs)	84.38	11.78	17.7	0.61×
BUSTER (1 GPU)	**86.30**	**0.89**	**496.7**	**16.6**×

B.8 Quantitative Evaluation of Burst-Time Explanations

We further evaluate burst-time explanations quantitatively on all 290 UCF-Crime test videos. Under a matched comparison of refinement policies, we measure whether the major anomaly event is visually supported by the evidence provided to the reasoning VLM. For ground-truth-overlapping explanations, we define event visibility as

$$\text{EventVis} = 100\left(1 - \frac{N_{\text{nv}}}{N_{\text{gt}}}\right), \tag{16}$$

where N_{nv} is the number of ground-truth-overlapping explanations whose major event is not visible in the selected evidence, and N_{gt} is the total number of ground-truth-overlapping explanations.

As shown in Table 16, Scout-and-Burst increases event visibility from 65.2% with Holmes-style allocation to 93.3%. It also avoids unnecessary anomaly explanations on normal videos: Holmes-style allocation generates explanations on 94 of 150 normal videos and produces anomalous interpretations on 39.3% of all normal videos, whereas Scout-and-Burst triggers none. Adding the same threshold gate to Holmes-style allocation removes these normal-video triggers, but its event visibility remains lower at 83.3%.

Table 16: **Controlled quantitative evaluation of burst-time explanations on UCF-Crime.** Normal Gen. denotes the number of normal videos on which an explanation is generated, and Normal Hall. denotes the percentage of all normal videos with an anomalous interpretation.

Policy	Event Vis. (%)	Normal Gen.	Normal Hall. (%)
Scout-and-Burst	**93.3**	**0/150**	**0.0**
Holmes-style	65.2	94/150	39.3
Holmes-style + θ gate	83.3	0/150	0.0

C Additional Qualitative Results

C.1 Additional Examples Across Standard datasets

In Figure 7, we provide additional qualitative examples to illustrate how the proposed scout-and-burst pipeline supports context-aware segment-level reasoning across different anomaly types and domains. In RoadAccidents141_x264 from UCF-Crime, the generated descriptions do not rely only on human-centered cues, but also capture the interaction between the motorcycle and the truck, the victim's post-impact state, and the reactions of nearby pedestrians. This suggests that the proposed system can ground anomaly reasoning not only in human actions but also in vehicle-object relations and broader accident context.

In the ShanghaiTech example 07_0047, the fighting sequence is short and visually less severe than many UCF-Crime samples, yet the model still identifies the aggressive interaction, the victim's response, and the surrounding observers. Interestingly, the comparatively lower anomaly score in this case appears consistent with the visual subtlety of the event, indicating that the score can reflect not only anomaly presence but also the strength and clarity of the observed evidence. This also suggests that stronger teacher-side priors or calibration could further improve sensitivity for such weak but semantically valid anomalies.

Finally, in the XD-Violence sample IpMan3, the generated descriptions capture not only the local violent act but also the progression of confrontation, the actors' reactions, and the escalation of threat across segments. The final burst segment, for example, explicitly identifies the knife threat and the resulting physical intimidation. Together, these examples suggest that efficient temporal sampling can preserve enough evidence for the reasoning VLM to recover object relations, situational context, and event progression across surveillance footage as well as more narrative movie-style clips.

C.2 Normal False Alarms

In Figure 8, we additionally highlight several normal samples on which the proposed system does not raise meaningful false alarms, despite conditions that are often challenging for motion-centric anomaly detectors. In Normal_Videos_310_x264, multiple people are gathered in a crowded basketball scene with frequent physical interaction and dynamic body motion. Such scenes can resemble fight-like motion patterns for detectors that rely heavily on motion dynamics, yet our model assigns essentially zero anomaly confidence and does not produce a misleading anomaly description. This suggests that the proposed framework does not simply react to crowdedness or physical activity, but instead uses the sampled evidence more selectively in context.

A similar pattern appears in Normal_Videos_882_x264, where the camera rotates around seated students in a classroom. Although abrupt camera motion and viewpoint change can

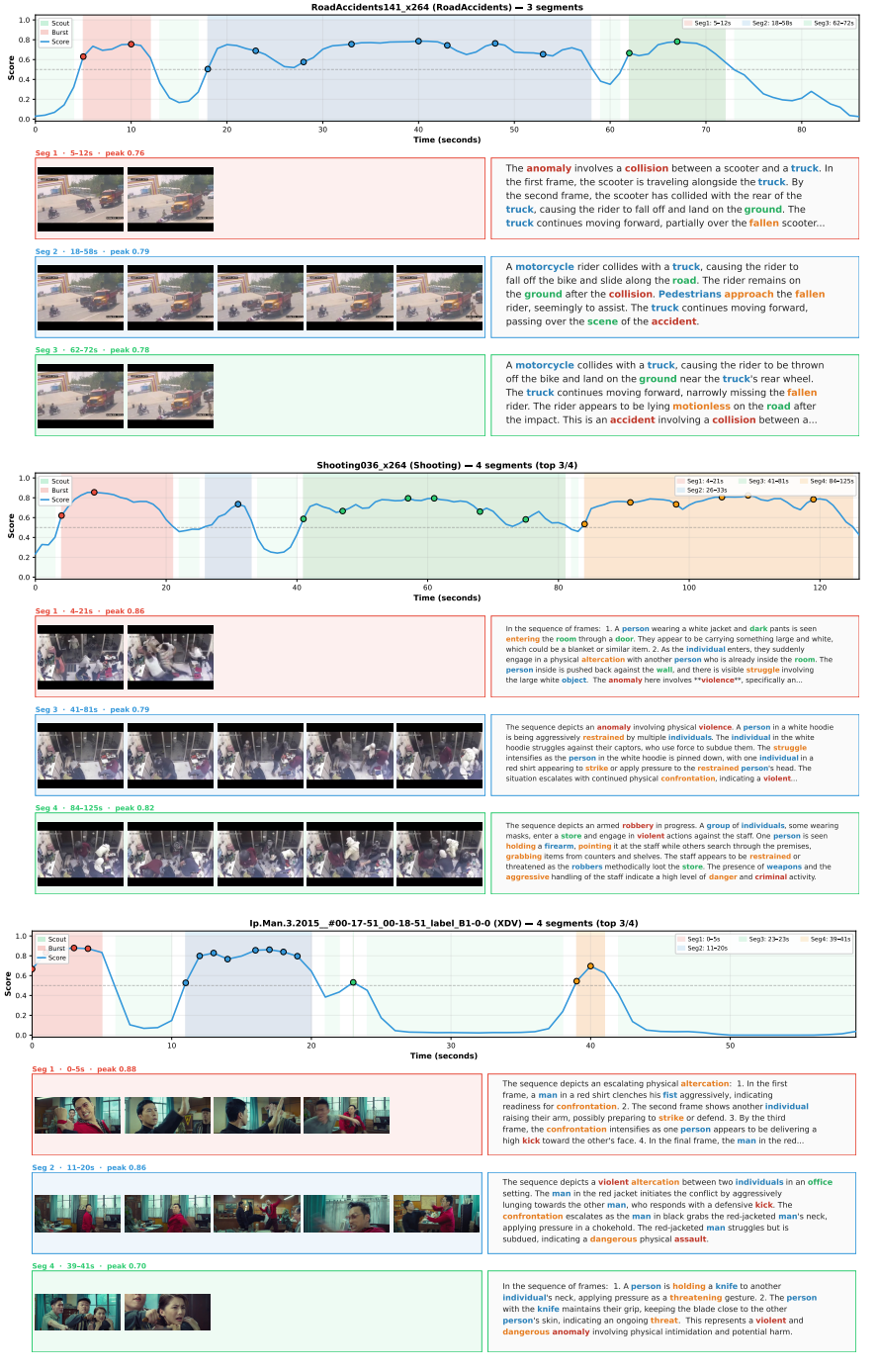


Figure 7: Qualitative scout-and-burst examples with on-demand VLM reasoning. Each row shows the score curve, burst-triggered segments, sampled frames, and segment-level descriptions. Color-coded words indicate objects, environment, interactions, and anomalies.

act as noise and trigger false alarms in motion-sensitive methods, the proposed system again keeps the sample in the normal regime. This indicates that the model is less sensitive to camera-induced motion artifacts and more robust to appearance changes that do not correspond to actual abnormal events.

A related observation can be made for the XD-Violence movie samples Before Sunrise and Good Will Hunting, where free human interaction alone does not induce a false alarm. Even in movie-domain clips with two actors engaged in natural conversation or social interaction, the model correctly describes the scenes as non-anomalous, without hallucinating violence, theft, accident, or other unusual activity. This suggests that the proposed system is not overly biased by prompt priors or pretrained knowledge toward anomalous interpretations whenever humans are present or interacting.

C.3 Failure Cases and Ambiguity Analysis

In Figure 9, we further analyze representative failure cases to better understand the limits of the proposed teacher-guided sampling and reasoning pipeline. These examples suggest that errors arise from at least two sources: perceptual difficulty in the visual evidence itself, and semantic mismatch between the dataset label and the most salient content described by the VLM.

In Abuse028_x264, the abnormal event is visually subtle and temporally brief: a postal worker appears to strike a dog through a fence, but the decisive action is short, partially occluded, and involves a very small target object. In this case, the model fails to extract the target action itself, indicating a teacher-side perceptual limitation when small-object motion and occlusion are combined. This example highlights a regime in which anomaly understanding can fail even before semantic reasoning becomes the main bottleneck.

In Shooting019_x264, the anomaly interval is localized correctly, but the generated explanation focuses on the visible post-event evidence, namely an incapacitated person lying on the ground near a vehicle, rather than on the shooting action that defines the class. This suggests that the model can detect that the scene is anomalous while still grounding its reasoning in a secondary consequence rather than the causally decisive interaction. In other words, the anomaly cue is captured, but the explanation is anchored to the aftermath instead of the action category itself.

A different failure mode appears in Shoplifting019_x264. Here, the model captures the relevant evidence well, including the store context, the cashier, and the aggressive human interaction, but interprets the event as a violent confrontation rather than as *shoplifting*. This case reflects a mismatch between the dataset label and the most visually explicit action in the clip: the sequence content is salient, but the class concept is semantically more ambiguous than the visible evidence suggests. A similar issue appears in 05_0017, where the model correctly identifies the key cue of throwing an object toward another person, yet maps it to *accident* rather than the dataset's anomaly label.

Taken together, these cases indicate that failure is not always caused by missing the anomalous interval itself. In some clips, the main difficulty is perceptual, such as small objects, short actions, or occlusion; in others, the anomaly is detected but described through a semantically adjacent interpretation, often because the dataset label is more specific, indirect, or ambiguous than the most visually grounded explanation. These examples motivate future work on teacher calibration, relation-aware reasoning, and more robust handling of subtle or conceptually ambiguous anomaly definitions.

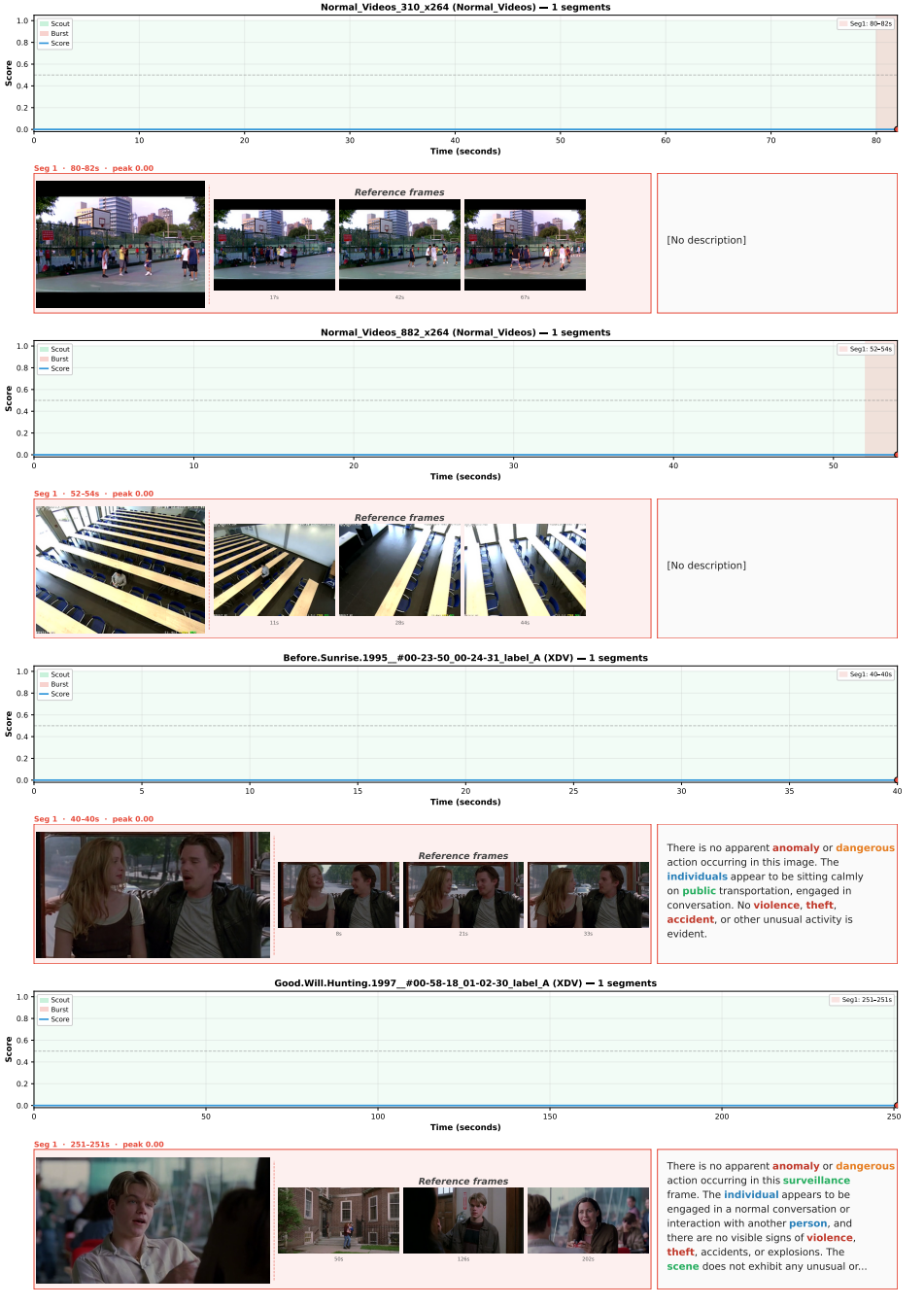


Figure 8: Qualitative 'Normal' examples with on-demand VLM reasoning. Each row shows the score curve, burst-triggered segments, sampled frames, and segment-level descriptions. Color-coded words indicate objects, environment, interactions, and anomalies.

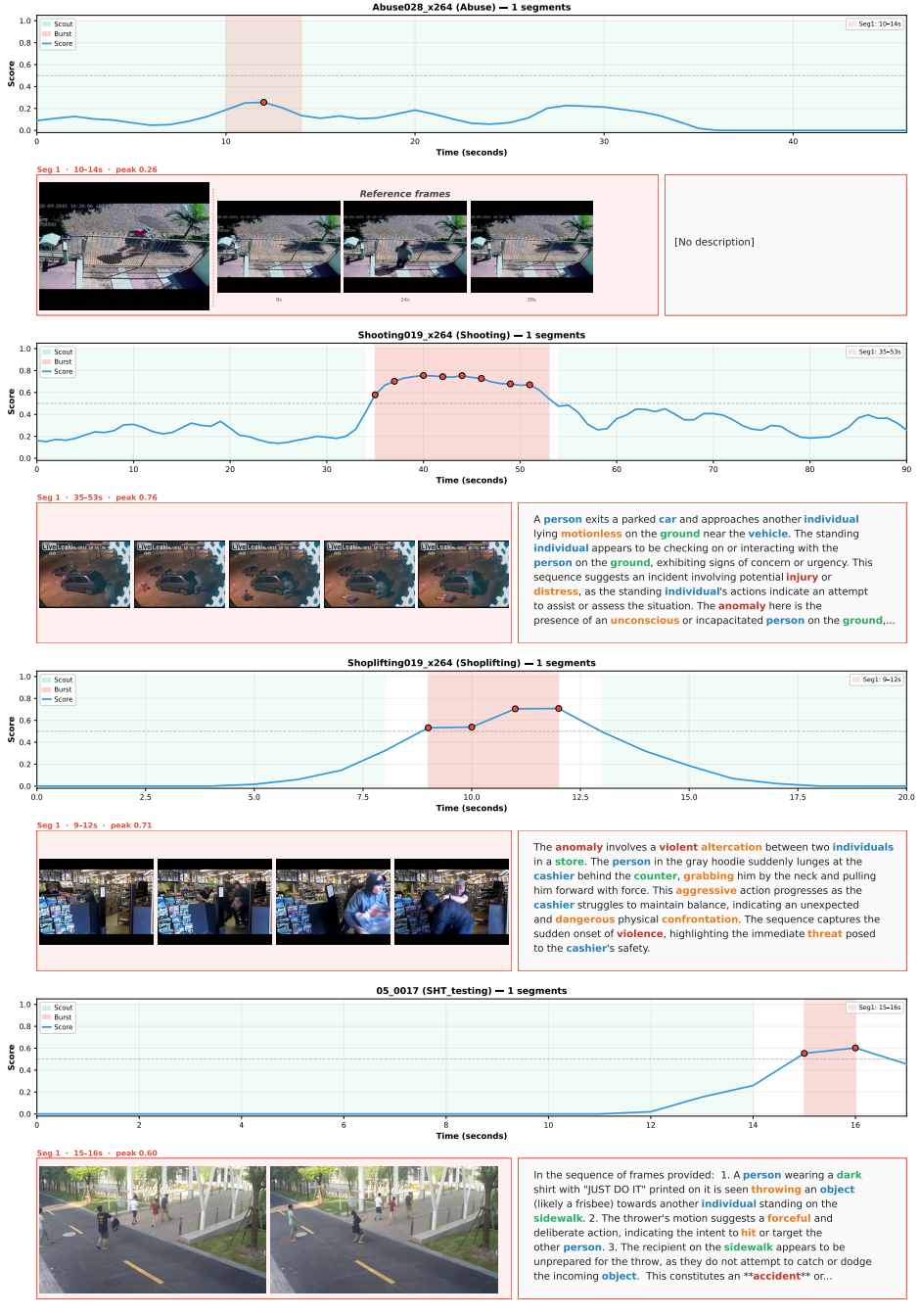


Figure 9: Qualitative 'Failure Cases' examples with on-demand VLM reasoning. Each row shows the score curve, burst-triggered segments, sampled frames, and segment-level descriptions. Color-coded words indicate objects, environment, interactions, and anomalies.